INCH-POUND

MIL-A-82843(OS)
5 February 1990
SUPERSEDING
WS 22380A
28 May 1986
(See 6.7)

MILITARY SPECIFICATION

ADHESIVE, ELEVATED TEMPERATURE CURING

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of elevated temperature curing, rubber-to-rubber and rubber-to-metal adhesive, referred to herein as the adhesive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-82823 Primer, Adhesive, Chlorinated Rubber Base

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1084 Viscosity of Adhesives (DoD adopted)

ASTM D 1875 Density of Adhesives in Fluid Form (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the adhesive shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Physical and mechanical properties.** The physical and mechanical properties shall be as specified in table I and 3.2.1.

TABLE I. Physical and mechanical properties.

Property	Limits	
	Minimum	Maximum
Viscosity at 77 ± 5°F, centipoise	450	850
Total solids, percent	22.5	26.5
Density, pounds per gallon	8.9	9.2

3.2.1 **Peel strength.** The adhesive shall have a peel strength of 80 pounds per inch of width when bonding ethylene propylene rubber (see 6.3) and steel using the adhesive specified herein and primer conforming to MIL-P-82823. Failure shall be not less than 90 percent cohesive in the rubber.

3.3 **Safety.**

3.3.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the

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appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2 and 6.5).

3.4 **Shelf life.** The adhesive shall have a shelf life of 12 months after date of shipment when stored at 80°F maximum.

3.5 **Workmanship.** The adhesive shall be uniform in appearance and shall be free from lumps, gels, and foreign material. No separation of components which is not readily dispersible shall be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. Quality conformance inspection.

Examination and Tests	Requirement	Test method
Visual examination	3.5 and Section 5	4.5.1
Viscosity	3.2	4.5.2
Total solids	3.2	4.5.3
Density	3.2	4.5.4
Peel strength	3.2.1	4.5.5

4.4.1 **Lot.** A lot shall consist of material which has been produced in one continuous operation under uniform manufacturing conditions, with no change in materials, and submitted for inspection at one time.

4.4.2 **Sampling.** Each lot of adhesive shall be randomly sampled to obtain the amount of adhesive necessary to perform all the tests specified herein.

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.5 and section 5.

4.5.2 **Viscosity test.** The viscosity of the adhesive shall be determined in accordance with ASTM D 1084, Method B, at 25 ± 1°C, using a Model LVF Brookfield viscometer, or equivalent, with a number 2 spindle at 30 revolutions per minute. Prior to performance of the viscosity test, the test sample shall be thoroughly mixed until all solids are in suspension. The viscosity of the material from each sampled container shall be recorded to the nearest centipoise.

4.5.3 **Solids content.** The solids content of the primer from each selected container shall be determined, using two test specimens, in accordance with the following:

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a. Add 0.8 to 1.2 grams (g) of sample weighed to the nearest 0.1 milligram (mg) into a tared aluminum weighing dish.

b. Coat sides of weighing dish with sample.

c. Place dish and contents into circulating-air oven at 154 ± 5°F for 2 hours ± 5 minutes.

d. Remove dish and contents from oven and place in a desiccator to cool.

e. When cooled to room temperature, reweigh dish and contents to nearest 0.1 mg.

f. Calculate solids content as follows:

$$\text{Solids content, percent} = \frac{C - A}{B} \times 100$$

Where: A = weight of dish, g

B = weight of sample, g

C = weight of dish and contents after heating, g

g. If the results of the initial two determinations differ by more than 0.5 percent, retest two additional samples.

h. Record the average of two determinations to nearest 0.1 percent.

4.5.4 **Density.** The density of the adhesive shall be determined using a weight per gallon cup in accordance with ASTM D 1875.

4.5.5 **Peel strength test.** The peel strength of the adhesive shall be determined on a 1-inch-wide bonded test specimen formed by vulcanizing a strip of rubber (see 6.3) to a strip of steel using the adhesive specified herein and primer conforming to MIL-P-82823 and dried for 60 to 90 minutes at 60 to 90°F. The cured specimen shall be placed in the jaws of a tensile apparatus and pulled at 45 degrees at a crosshead speed of 20 inches per minute. The force required to separate the bond and the mode of failure shall be recorded.

4.6 **Inspection of packaging.** The adhesive containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

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5.1.1 **Level C.** The adhesive shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Manufacturer's name and address.

b. Date of manufacture.

c. Shelf life expiration date.

d. Storage conditions.

e. Title, number and date of this specification.

f. Toxicity warning, if applicable.

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The adhesive is intended for use in rubber-to-rubber and rubber-to-metal bonds in the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article inspection is required (see 3.1).

d. Material Safety Data Sheets Required (see 3.3.2 and 6.5).

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e. Special marking, if other than as specified in 5.2.2.

6.3 **Ethylene propylene rubber formulation.** The rubber used for the peel test specified in 4.5.5 should be ethylene propylene rubber of the following formulation (HC-705 manufactured by Lord Corporation, Chemical Products Group is a typical product):

Ingredients	Parts by Weight
a. Masterbatch:	
(1) Vistalon 4608	100.0
(2) Zinc oxide (Protox 169)	5.0
(3) Stearic acid	1.0
(4) HAF Black (United 60)	80.0
(5) Flexon 765	40.0
	226.0
b. Accelerator and Vulcanizing Agents:	
(1) Sulfur	1.5
(2) Thionex	1.5
(3) Captax (MBT)	0.5
	3.5

6.4 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

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6.6 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.7 **Supersedure information.** MIL-A-82843 incorporates the following engineering change proposals (ECPs) and specification change notices (SCNs):

ECP	SCN
	SCN 3 (12/5/86)
MTA005 (3/25/87)	
MTA046 (1/21/88)	SCN 4 (2/8/88)

Preparing Activity:
NAVY-OS
(Project 8040-N155)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82843	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
ADHESIVE, ELEVATED TEMPERATURE CURING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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